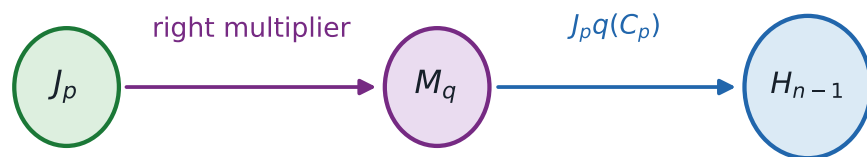


Residue duality reaches a native moment Gram through a typed multiplier

Native monic orthogonal polynomial p_n



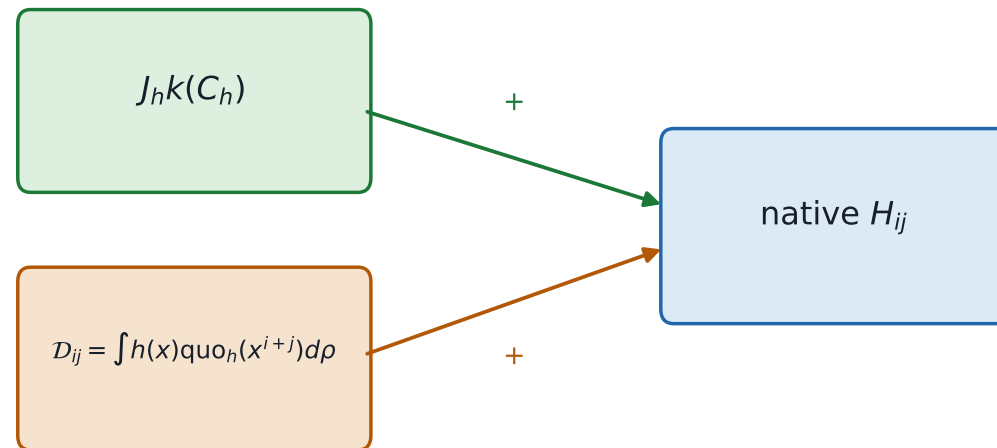
$$q(z) = \int \frac{p_n(z) - p_n(x)}{z - x} d\rho(x)$$

leading coefficient = μ_0 ; $q(C_p)$ is a unit

$$\int \frac{d\rho}{x+a} + \frac{q(-a)}{p_n(-a)} = \frac{1}{p_n(-a)^2} \int \frac{p_n(x)^2}{x+a} d\rho \geq 0$$

hypothesis: $\text{supp } \rho \subset [0, \infty)$, $a > 0$

Arbitrary monic ES root polynomial h



The defect vanishes when $h = p_n$ by orthogonality.
For a generic ES quartic it remains part of the exact map.